

Licensing Scheme

Unlike direct licensing schemes which require annual payment and at times royalties as well, the HyperTransport licensing strategy is comparatively simplified. Those who want to use the technology may join the consortium for \$5,000 a year, and those who also want to contribute to the technology development may join for \$25,000 a year. There are no royalties or licensing fees beyond this annual membership cost. At the moment, licensing information on 3GIO is not available.

pair can carry 1.6 Gbps (200 MBps) of data. Other HyperTransport implementations will be using a slower but cheaper HyperTransport IO system, at a speed of 800 Mbps (100 MBps) per pin pair, which is just under the speed of a 32-bit PCI.

HyperTransport is a differential bus. This means that for each bit of the connection, two wires (and therefore interface pins on a chip) are required to carry the signal. Wire one carries the signal and wire two carries the inverse of the signal. For each device residing on a bus, there needs to be a corresponding number of pins. This increases the pin count by quite a margin, so many implementations of this technology will use reduced bus widths to cut costs.

HyperTransport products

Products from AMD based upon the HyperTransport protocol are expected to appear in 2002. HyperTransport has the potential to replace most point-to-point busses in use in computer systems today. This includes north bridge-south bridge interconnects, the memory bus, AGP, and the front-side-bus. AGP is currently limited to 4x speed, and 8x speed is not an easy task. With nVidia licensing HyperTransport for their own use, both within their chipsets and between their chipsets and other components, the appearance of a HyperTransport graphics processor (GPU) is guaranteed.

This raises the interesting spectre of a system without any AGP slots, and a lot more integration onto the motherboard. Now that nVidia are into chipset design, they could very well decide to drop AGP support and make the GPU part of that chipset, connected using HT to the north bridge at 1.6 GBps. The standard specification would mean that any HTIO GPUs that ATI or Matrox produced would also work with this chipset, meaning more choice for the consumer.

3GIO

Intel's latest initiative to develop a standardised bus transport comes in the form of third-generation I/O, or 3GIO for short. Like HyperTransport, 3GIO (code named Arapahoe) addresses similar concerns of bandwidth, scalability and flexibility in design.

3GIO is a point-to-point interconnect and uses a packetised protocol structure. It transports data at high bandwidths, uses scalable bus widths, exhibits low power consumption and is hot-pluggable. It was designed for chip-level interconnects, expansion card connections and can act as an I/O bridge for other interconnects such as USB 2.0, Infiniband, Ethernet, 1394, etc. 3GIO is capable of providing interlinks within the chipset core but initial proposals state that initial implementations will focus on the I/O devices and subsequently the AGP bus itself.

The specific workings of 3GIO differ somewhat from HyperTransport. A fundamental 3GIO bus link uses two low-voltage pairs of signals, one pair to transmit and the other to receive. Initial speeds between pairs is over 300 MBps and this is expected to increase to 1.25 GBps (the physical limits for copper interconnects). Once these limits are reached, 3GIO has made provisions for a smooth transition from copper to faster interconnects using optical links.

3GIO also brings forth a technique to allow itself to connect to external high-speed devices such as hard disks, optical drives and video cameras. Furthermore, 3GIO can be used to tie together different PC subsystems that are housed in physically separated enclosures. This in itself could be used to completely redesign the 'beige box' existence of PCs

